HAD6750 Lecture 6: Competition and Health Systems

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1 Gaynor & Town (2012); Gowrisankaran & Town (1997)

Structural models allow us to answer a specific set of questions:

1. How do providers/hospitals choose when to enter/exit a market? Innovate? Invest? **Merge?** [note that merger analysis is a huge place where we use structural models]
2. How does imperfect competition change welfare outcomes?
3. How does health policy affect market structure?

Structural models are unique because they have multiple agents each optimizing simultaneously (and hence, endogenously):

- In each period, there are $s_t \in S$ **participants**
- Each participant has a set of **actions** $a_t \in A$ and a corresponding **payoff** function $u_t(a_t; s_t, \xi_t)$,
- where ξ_t corresponds to agent-specific idiosyncratic unobservables (**shocks**)
- Models can be *static* (one-period) or *dynamic*, where there are state transitions across periods: $s_{t+1} = f(s_t, a_t, \xi_t)$. [*Note*: can you explain why this might depend on each of its arguments?]

Example of a Dynamic Problem: Hospital Entry

Suppose that we consider a set of hospitals that are *forward-looking*, so that they maximize expected profits over the future:

$$\max_{\{a_t\}_{t=1}^{\infty}} \mathbb{E}\left[\sum_{t=1}^{\infty} \beta^t u_t(s_t, a_t, \xi_t)\right] \quad \text{s.t.} \quad s_{t+1} = f(s_t, a_t, \xi_t).$$

Suppose that in the model, $a_t \in \{\text{enter}, \text{exit}, \text{merge}, \text{invest}\}$. Which of these are extensive margin effects? Which of these are intensive margin effects? [For now, let's focus on extensive margin.]

Model Timeline

1. First, each hospital draws **scrap values**, private valuation of exiting:

$$s_{jt}^{\text{exit}} \sim U(\phi - \sigma_\phi, \phi + \sigma_\phi).$$

2. If we want the intensive margin, each hospital might also draw **investment values**, driving valuation of investing further

- To model this, suppose that there is an "efficiency" parameter $\omega_{jt}(x_{jt}) \in \xi_{jt}$

3. Third, hospitals out of the market make **entry decisions** [note that the timing here is important!]
$$s_{jt}^{\text{entry}} \sim U(s - \sigma_s, s + \sigma_s),$$
with a starting investment ω_0.

But where are all these parameters coming from? What's important to include, and what can be left out? What is homogeneous, what is time varying, what is known and what is unknown/random?

Firm Objective Function

Firms try to maximize profits:

$$\max \pi = D(w)P - c(w) - c(x) + \xi + 1\{s_{jt}^{\text{entry}}\} + 1\{s_{jt}^{\text{exit}}\}.$$

1. What does it look like to estimate this model?
2. What does it look like to use it for counterfactuals?
3. What assumptions can we make for this to be a simpler model? What can we do to make it more complex?

Now let's think about how to incorporate an additional stage: **insurer contracts**. This is better done in the slides.

2 Gaynor, Ho & Town (2015)

Finally, let's enrich the model a bit further by contemplating how **patients** choose hospitals given **hospital** and **insurer** behavior. Why would we want this? Merger analysis that lets us take into account market power!
Model timing:

1. Insurers (H plans) and hospitals (N hospitals) bargain over prices and inclusion in networks
2. Patients (I patients) choose health plans/networks
3. Patients realize illness shocks, seek care based on results in (1) and (2); hospitals and insurers realize profits.

What do agents know about each other?

- "Passive beliefs": competitors are acting independently (offer from insurer 1 gives me no information about insurer 2)
- Health shocks and patient choices are well-understood (ha!)

How are we going to solve a model this complicated? **Backward induction**! So we start with step 3:

Step 3: Patient choice

Once patients have enrolled in a plan, they face shocks—at this point, these are **known**. Suppose patients face health shocks $m \in \{0, m_1, ..., m_M\}$ with probability ρ_m and severity weight w_m [what are these?].

Patients can then seek care in-network (normalize OOP= 0), out-of-network (higher prices), or avoid care. Which option can we ignore in the model, and why? Do we want to do this?
Patient utility is therefore:

$$u_{ijm} = w_m \left[F(x_j, z_i, d_{ij}, m_i; \theta) + \varepsilon_{ijm} \right].$$

That is,

- Utility depends on hospital, patient, and illness characteristics, distance to hospital.
- θ is the vector of structural parameters to be estimated.
- We assume that ε_{ijm} is Type 1 Extreme Value.
- In practice, u is parameterized to be a linear function of a hospital fixed effect, travel time and interactions between hospital characteristics (such as bed size, ownership type, teaching status, service offerings), patient characteristics, diagnosis, diagnosis weight, and travel time.

Given these assumptions, we can derive **choice probabilities**, the probability that patient i chooses hospital j in their network:

$$\Pr(i \text{ chooses } j|m) = \frac{\exp(w_m f(i,j,m)}{1 + \sum_k \exp(w_m f(i,j,m))}.$$

Note that the denominator is the **outside option** (1) plus the sum of all other possible utilities.

Step 2: What plans do consumers choose

Plans have to be chosen **before** health shocks are realized. If the shocks are unknown, then we can write the *ex-ante* expected utility as:

$$W_i(J_h) = \sum_m \left(\rho_{i,m} \, w_m \, \ln \left(1 + \sum_{j \in \mathcal{J}_h} \exp(f(i,j,m)) \right) \right).$$

Note that W changes with our parameters, θ; patients pick J_h to maximize $(W|\theta)$ [FOCs omitted, can you derive them?]

- Slides have some points about empirical estimation here

Step 1: Nash bargaining

Given that hospitals and insurers know how patients will respond to their plan features, how do they design the networks and prices?

First, consider **insurance plans'** utility:

$$\pi_h(J, \xi, Prem) = (Prem_h - ac_h - \sum_j p_{jh} q_{jh}) Y_h - fc_h,$$

where we have:

- $Prem_h$: premiums need to cover expected payouts ($\sum_j p$) and overhead (variable) costs ($-ac_h$). [What are we summing over here and why?]
- Y_h: market share of each plan, across populations
- fc_h: why don't we need to worry about fixed costs?

When firms maximize these profits, their first order condition is:

$$\frac{\partial \pi_h}{\partial Prem_h} \Rightarrow Y_h + (Prem_h - ac_h - \sum_j p_{jh} q_{jh}) \frac{\partial Y_h}{\partial Prem_h} = 0.$$

- What must be satisfied in equilibrium? What does this mean?
- Are we sure that an equilibrium exists? **No!** It's not a trivial thing given how complicated the model has become. In this case, Caplin & Nalebuff (1991) prove that there is a unique equilibrium in the premium setting game given the set of health plan hospital networks. The premiums and thus the profits of all the health plans will depend upon the structure of all the competing plans' hospital networks, their costs, and the distribution of patients across geography and incomes.

Contracts

Finally, we can discuss insurer-hospital contracts. We assume:

- Assume that contracts are only over prices of inpatient care p_{jhm}
- We can use one price p_{jh} weighted by w_m for each m
- Hence, health plan will pay $w_m \, p_{jh}$ for any realized health shock m
- This is a big simplifying assumption! But there are still $J \times H$ contracts we need to consider!

So what does negotiation look like? There are multiple options, but we consider **Nash baragaining** with the following timing:

1. Look at each pair in isolation
2. Each pair gets a "match quality" cm_{jh} (another random unobserved variable)
3. One player (random) makes first offer according to their **agreement values** (below)
4. Ignores strategic interactions across hospitals/payers (what does this miss?)

Hospitals want to maximize revenue:

$$F_h(J_h) = Prem_h^*(J_h, J_{-h}) \times Y_h(J_h, \xi, Prem_h^*)$$

Hence, insurer disagreement value is:

$$F_h(J_{h-j}) = \underbrace{\left[Prem_h^*(J_{h-j}, J_{-h}) \times Y_h(J_{h-j}, \xi, Prem_h^*)\right]}_{\text{insurer loss in coverage of hospital } j} - \underbrace{\sum_{k \in H} p_{kj} \, q_{kh}^{J_{h-j}}}_{\text{patient flows over hospitals w/in networks } h}$$

Note that disagreement is helpful for one party (loss in value of excluding hospital j from network). Hence, in Nash equilibrium, the party that moves first maximizes their agreement value taking into account the disagreement value of the other party (and attempting to find a **best-response** function). If β represents the **insurer's** bargaining power, then bargaining chooses price to maximize:

$$\max_{p_{hj}} \left(H_{agree} - H_{disagree}\right)^{\beta} \left(M_{agree} - M_{disagree}\right)^{1-\beta}.$$

This has the first order condition:

$$p_{jh} = (1-\beta)\left(c_j - \frac{r_j}{q_{jh}}\right) + \beta(F_h(J_h) - F_h(J_{h-j}) - c\, m_{jh}) + \beta \sum_{\ell \neq j} p_{\ell h}\, d_{j\ell h},$$

where $d_{j\ell h}$ indicate patient shares (which we indirectly solved for above).

How do we solve this model? Better to do it empirically than theoretically from here:

1. Estimate patient demand for hospitals: logit model of demand (IO classic). Generally, this requires (1) simulating shocks for all random variables in the model; (2) iterating over guesses of theta; until (3) some kind of convergence.
2. Estimate value of adding hospital to network based on ex-ante utilities from (1)
3. Estimate hospital/insurer surplus and bargaining power using additional data on hospital revenues and costs.
4. Once this is done, you can estimate the parameters in the bargaining FOC
5. Then you can calculate merger counterfactuals with predicted post-merger prices, and run counterfactual exercises based on changes to structural parameters

Then you can take the solution and answer questions such as:

1. What are the comparative statics here?
2. How do mergers change this problem?
3. What structural (counterfactual) exercises can we use this model to simulate?
4. What types of welfare can we measure?

For example, we can directly measure **consumer surplus** as:

$$CS_i\big(Prem_h, W_i, c_h, \xi_h; \alpha^P, \gamma, \theta\big) = \frac{1}{\alpha^P} \ln\left[1 + \sum_h \exp(v_{ih}^*)\right],$$

taking into account the optimal premiums at the patient level.